Experiment 14

Construct a C program to organise the file using a single level directory.

C Program

```
#include <stdio.h>
#include <string.h>

#define MAX 20
#define NAME 30

int main() {
    char files[MAX][NAME];
    int count = 0, choice;
    char name[NAME];

    while (1) {
        printf("\n1. Create File\n2. Search File\n3. Display Files\n4. Delete File\n5. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        if (choice == 1) {
            if (count == MAX) {
                printf("Directory is full.\n");
                continue;
            }
            printf("Enter file name: ");
            scanf("%29s", name);

            int exists = 0;
            for (int i = 0; i < count; i++)
                if (strcmp(files[i], name) == 0) exists = 1;

            if (exists)
                printf("File already exists.\n");
            else {
                strcpy(files[count++], name);
                printf("File created successfully.\n");
            }
        } else if (choice == 2) {
            printf("Enter file name to search: ");
            scanf("%29s", name);
            int found = 0;
            for (int i = 0; i < count; i++)
                if (strcmp(files[i], name) == 0) found = 1;
            printf(found ? "File found.\n" : "File not found.\n");
        } else if (choice == 3) {
            printf("Files in single level directory:\n");
            for (int i = 0; i < count; i++)
                printf("%s\n", files[i]);
        } else if (choice == 4) {
            printf("Enter file name to delete: ");
            scanf("%29s", name);
            int pos = -1;
            for (int i = 0; i < count; i++)
                if (strcmp(files[i], name) == 0) pos = i;

            if (pos == -1) printf("File not found.\n");
            else {
                for (int i = pos; i < count - 1; i++)
                    strcpy(files[i], files[i + 1]);
                count--;
                printf("File deleted successfully.\n");
            }
        } else if (choice == 5) {
            break;
        } else {
            printf("Invalid choice.\n");
        }
    }

    return 0;
}
```

Sample Output

```
1. Create File
2. Search File
3. Display Files
4. Delete File
5. Exit
Enter choice: 1
Enter file name: data.txt
File created successfully.

Enter choice: 3
Files in single level directory:
data.txt
```

Result

The single-level directory structure was successfully simulated with create, search, display and delete operations.